Assignment No 1:

Q1. Number System Conversions
a) Convert $(756.34)_8$ to decimal.
b) Convert $(245)_{10}$ to binary and hexadecimal.
c) Convert $(101101.101)_2$ to decimal.
d) Convert $(3A7)_{16}$ to binary and decimal.

Question no 2

Q2. Signed Numbers & Complement System
(a) Complement Basics
Find 1's and 2's complement of:
(i) 10101100
(ii) 01101010
Represent the following numbers in 8-bit 2's complement form:
(i) -45
(ii) -92
(iii) +78
(b) Subtraction Using 2's Complement Method
Perform the following using 8-bit system:
34 − 19
19 − 34
-25 − 17
45 − (-18)

Q3
Binary Fraction Conversions
Convert $(110101.1011)_2$ to decimal.
Convert $(0.375)_{10}$ to binary and verify by converting back to decimal.
Convert $(1011.011)_2$ to octal.

Q4:
Octal Addition (Base 8)

(56) base 8 + (27) base 8
(135) base 8 + (247) base 8
(672) base 8 + (354) base 8
(507) base 8 + (468) base 8
(777) base 8 + (1) base 8
(645) base 8 + (732) base 8

Hexadecimal Addition (Base 16)

(A=10, B=11, C=12, D=13, E=14, F=15 use karein)

(A7) base 16 + (3D) base 16
(5F) base 16 + (8C) base 16
(9B) base 16 + (7E) base 16
(2AF) base 16 + (19C) base 16
(FFF) base 16 + (1) base 16
(ABC) base 16 + (DEF) base 16

Octal & Hexadecimal Subtraction

Part A: Octal Subtraction (Base 8)
(65) base 8 − (27) base 8
(154) base 8 − (67) base 8
(703) base 8 − (256) base 8
(1000) base 8 − (1) base 8
(742) base 8 − (365) base 8
(600) base 8 − (457) base 8
Part B: Hexadecimal Subtraction (Base 16)
(Note: A=10, B=11, C=12, D=13, E=14, F=15)
(A5) base 16 − (3C) base 16
(7F) base 16 − (29) base 16
(1B4) base 16 − (9D) base 16
(200) base 16 − (1F) base 16
(ABC) base 16 − (5E7) base 16
(1000) base 16 − (1) base 16

Q5
Binary Subtraction
1000 − 0011
10100 − 00101
11010 − 01011
10000 − 01101
11101 − 10111